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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

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*2044705592*

**MATHEMATICS (SYLLABUS D)** **4024/12**

Paper 1 **October/November 2010**

**2 hours**

Candidates answer on the Question Paper.

Additional Materials: Geometrical instruments

**READ THESE INSTRUCTIONS FIRST**

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

If working is needed for any question it must be shown in the space below that question.
Omission of essential working will result in loss of marks.

**NEITHER ELECTRONIC CALCULATORS NOR MATHEMATICAL TABLES MAY BE USED IN THIS PAPER.**

The number of marks is given in brackets [ ] at the end of each question or part question.
The total of the marks for this paper is 80.

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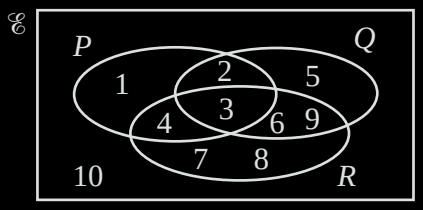
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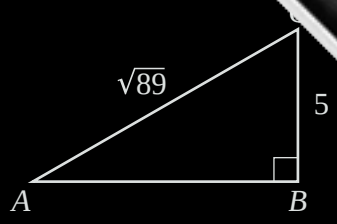



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$\sqrt{\phantom{x}}$

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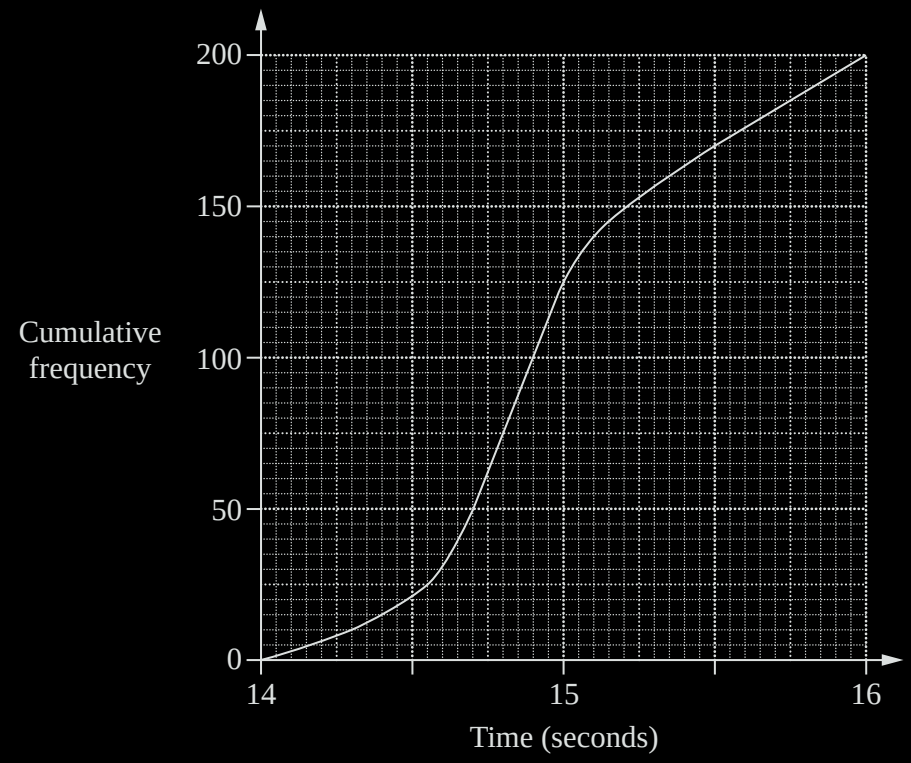


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Use the curve to find

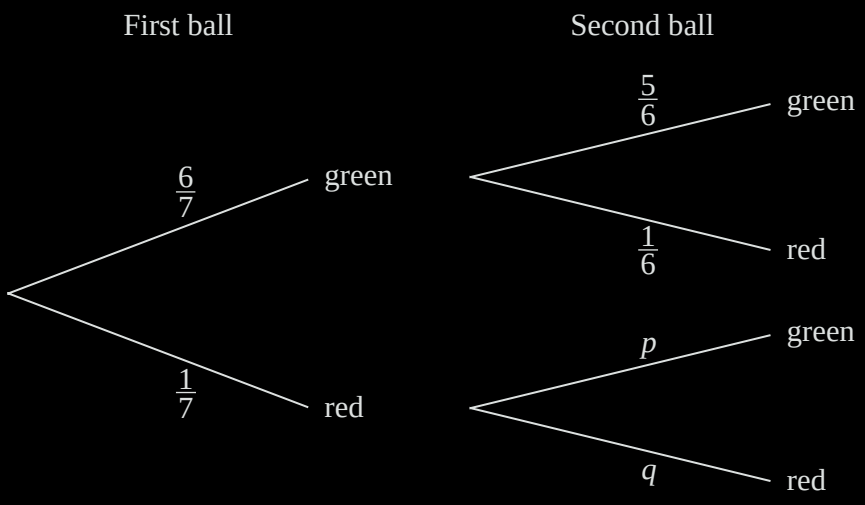
**(a)** the lower quartile,

*Answer* (*a*) ..... seconds [1]

**(b)** the number of children who took at least 15.5 seconds.

*Answer* (*b*) .....[2]

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**(a)** Find the values of $p$ and $q$.

*Answer* (*a*) $p$ = ..... $q$ = .....[1]

**(b)** Expressing each answer as a fraction in its simplest form, find the probability that

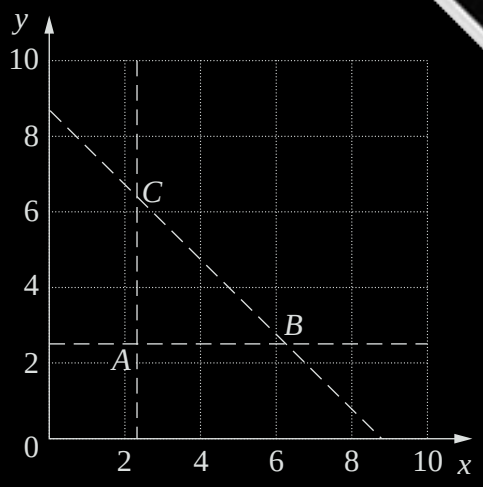
**(i)** both balls are green,

*Answer* (*b*)(i) .....[1]

**(ii)** the two balls have different colours.

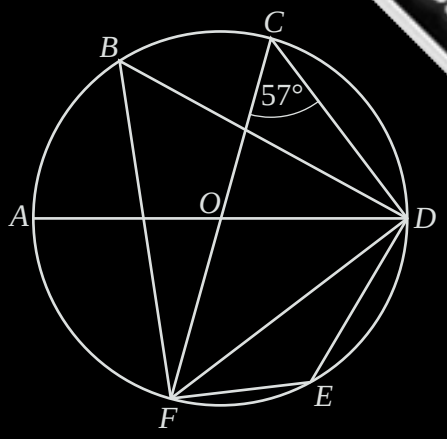
*Answer* (*b*)(ii) .....[1]

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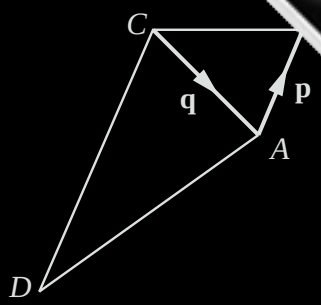


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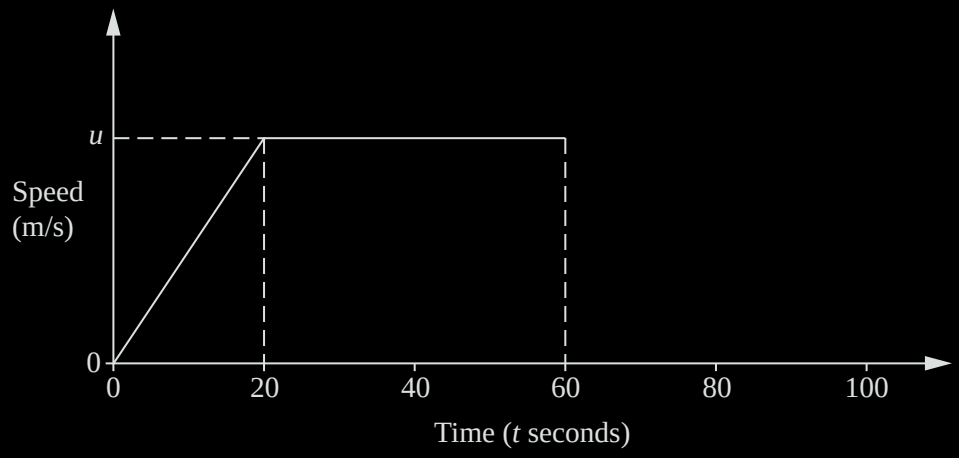
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The diagram is the speed-time graph of part of the journey of a car.
From $t = 0$ to $t = 20$ the car moves with a constant acceleration.
From $t = 20$ to $t = 60$ the car moves with a constant speed of $u$ metres per second.

**(a)** When $t = 20$ the car has travelled $D$ metres from the start.

Calculate the value of $t$ when the car has travelled $2D$ metres **from the start**.

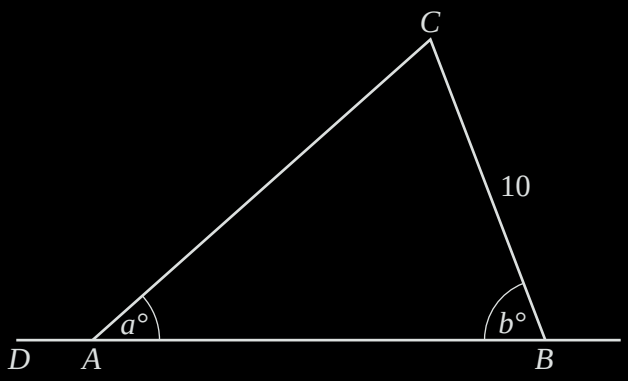
*Answer* (*a*) $t =$ .....[2]

**(b)** At $t = 60$, the car slows down with a constant deceleration.
This deceleration is half of the acceleration between $t = 0$ and $t = 20$.

During this period of deceleration, calculate the value of $t$ when the car has a speed of $\frac{u}{4}$ metres per second.

*Answer* (*b*) $t =$ .....[2]

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| $x°$ | $a°$ | $b°$ |
|---|---|---|
| $\sin x°$ | $\frac{3}{5}$ | $\frac{24}{25}$ |
| $\cos x°$ | $\frac{4}{5}$ | $\frac{7}{25}$ |
| $\tan x°$ | $\frac{3}{4}$ | $\frac{24}{7}$ |

In the diagram, $DAB$ is a straight line.
$BC = 10$ cm, $C\hat{A}B = a°$ and $C\hat{B}A = b°$.

Use as much information given in the table as is necessary to answer the following questions.

**(a)** Write down the value of $\cos D\hat{A}C$.

*Answer* (*a*) .....[1]

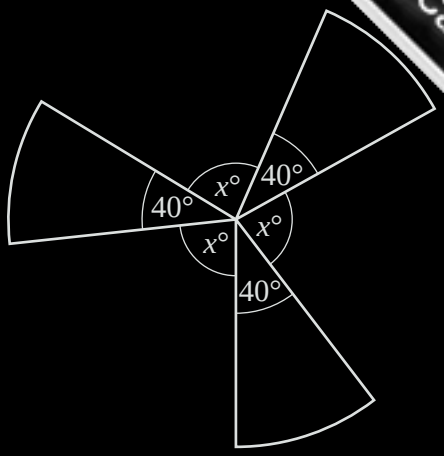
**(b)** Calculate $AC$.

*Answer* (*b*) $AC =$ ..... cm [3]

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**Question 27 is printed on the following page**

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